WATERSHED MODELING

USPCASW, MUET

FALL 2026

RUNOFF MODELS

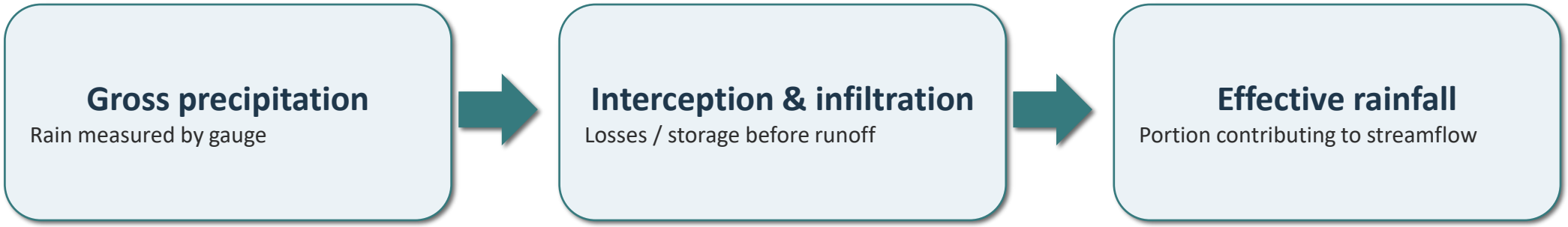
LEARNING OBJECTIVES

- From rainfall to runoff — effective rainfall, infiltration and runoff generation
- Hydrograph analysis — rising limb, peak and recession, and what controls hydrograph shape
- Rainfall–runoff modeling — inputs, method choice, calibration, validation and trust
- Two families of runoff models — peak runoff models and continuous runoff models
- The Rational Method — assumptions, the peak-

1. From rainfall to runoff

Effective rainfall, infiltration and runoff generation

From Gross Rainfall to Effective Rainfall



Effective precipitation = gross precipitation – losses

- Effective rainfall is the portion of rainfall that produces a perceptible streamflow response.
- Not all rainfall becomes surface runoff; some infiltrates, is stored, or contributes to subsurface flow.

Infiltration and Runoff Generation

- Dry soil may initially have a high infiltration capacity.
- If rainfall intensity/rate is below infiltration capacity, little or no overland flow occurs, although interflow may still develop.
- With continuing rainfall, soil becomes wetter, and infiltration capacity can decline.
- When rainfall exceeds the capacity of the soil to absorb water, excess water contributes to runoff.

Rainfall–Runoff Response

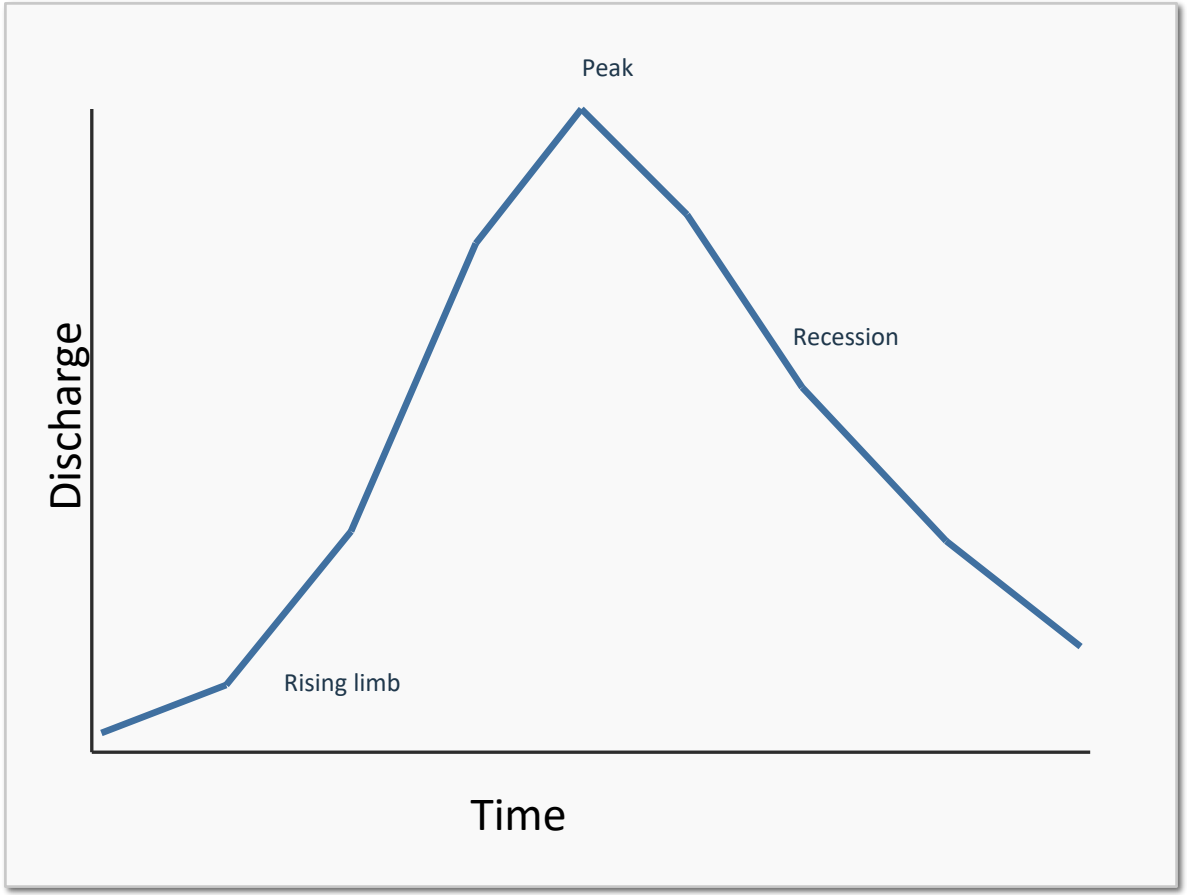
- A catchment can be treated as a system that transforms rainfall input into streamflow output.
- Inputs include rainfall amount, intensity, duration, and spatial distribution.
- The catchment controls the transformation through soils, land cover, slope, drainage, and storage.
- The observed output is discharge at the catchment outlet.

2. Hydrograph Analysis

From observed rainfall to streamflow

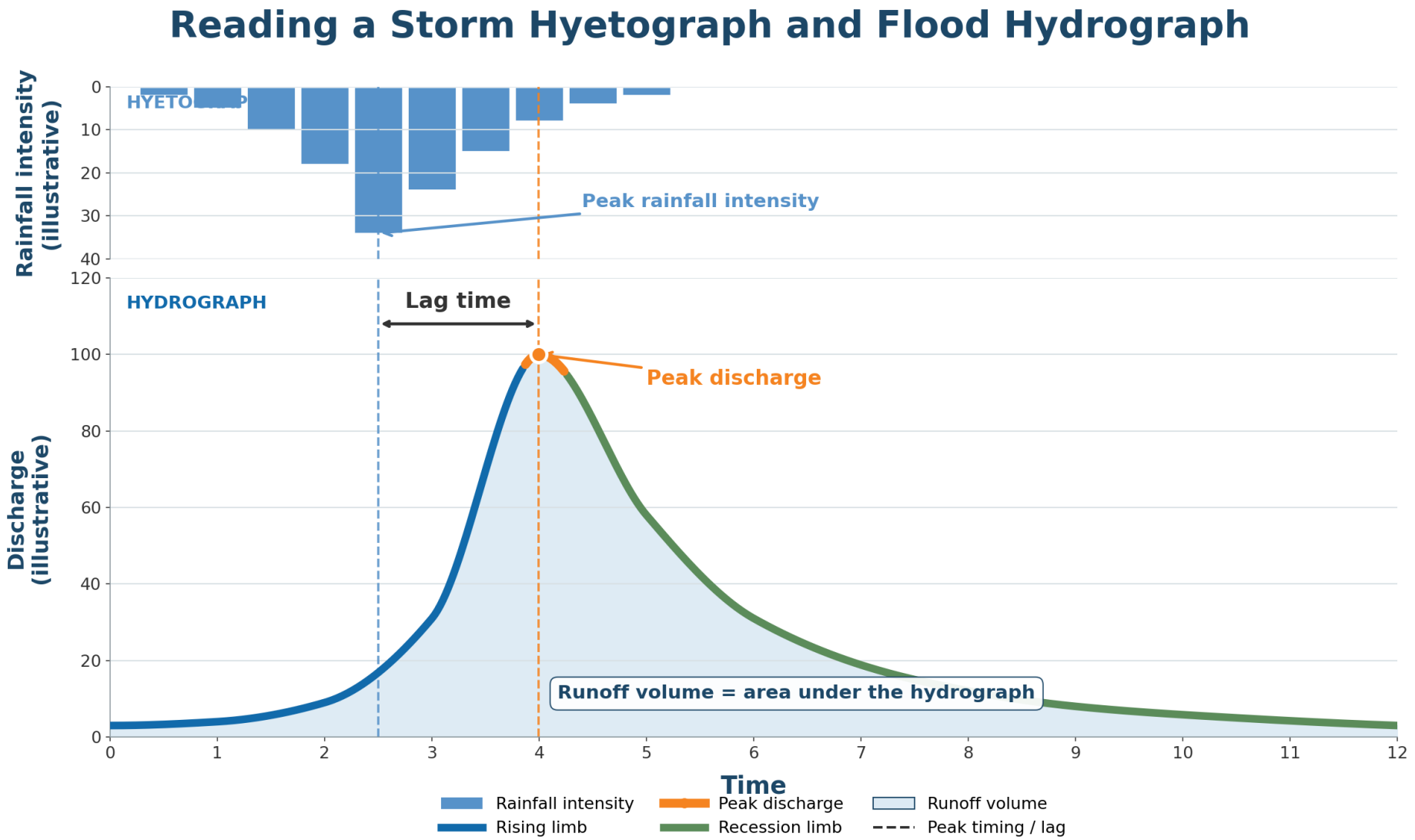
Hydrograph

- **Hydrograph:** discharge plotted against time at a channel location.



How do we read a Flood Hydrograph?

Peak, timing, and volume—all matter for design.

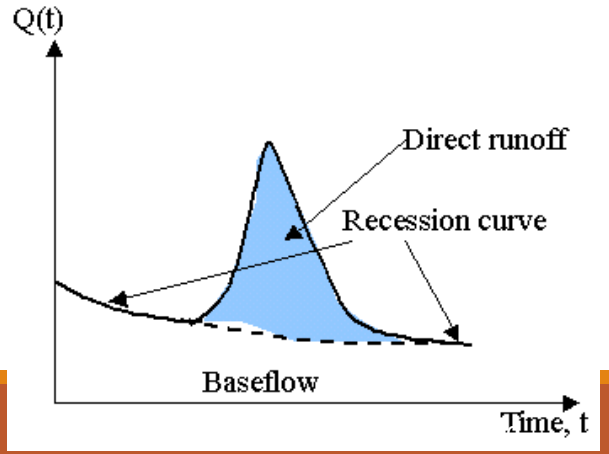


Hydrograph Analysis

- Hydrologic response of rainfall at the outlet of an area
- Hydrograph: Graph of discharge (cubic feet per second) in a channel vs. time
- Area under curve yields the volume of runoff

Stream flow = Direct Runoff + Base flow

Direct Runoff (DRO) = Rainfall Excess or (rainfall – losses)



Source: Prof. Ke-Sheng Cheng

What controls hydrograph shape?

Storm/climate

- Rainfall Intensity
- Rainfall Duration
- Rainfall Spatial Distribution
- Rainfall Temporal Distribution

Catchment

- Catchment Size and Shape
- Slope and Elevation
- Soil and Land cover
- Drainage density, Roughness, and Storage
- Antecedent Moisture

How Catchment Characteristics Change Response?

- Steeper slopes generally produce faster runoff and a sharper hydrograph response.
- Impervious or highly developed surfaces can reduce infiltration and increase rapid runoff.
- Larger catchments generally produce greater runoff volume for the same rainfall depth, but their response is often broader and slower.
- Vegetation, soil properties, channel length, drainage density, and natural/channel storage also modify the response.
- Snowmelt at high elevation can produce broader, lower-peaked flood responses than intense rainfall.

"Why Does the Same Rainfall Produce Different Runoff?



Watershed A	Watershed B
Gentle slopes	Steep slopes
Permeable soils	Shallow or less permeable soils
More vegetation	Sparse vegetation
More water soaks into the ground	More water flows over the surface
Less and slower runoff	**More and faster runoff**

Key message: Rainfall alone does not determine runoff.
Soil, slope, vegetation, and existing moisture conditions also matter.

Different watershed characteristics

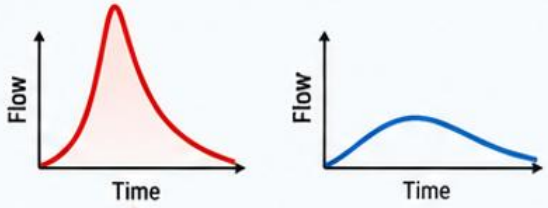
Same rainfall
(e.g. 50 mm)

Different runoff response

1. Slope (Topography)

Steeper slopes generate faster and higher runoff.



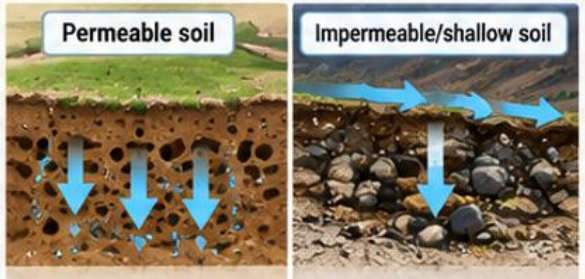


Higher and earlier peak

Lower and later peak

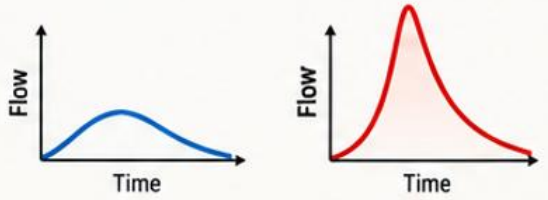
2. Soil and Infiltration

Permeable soils allow more water to soak in, reducing runoff.



More infiltration
Less runoff

Less infiltration
More runoff

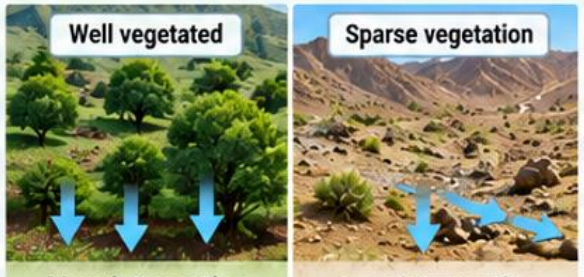


Lower peak

Higher peak

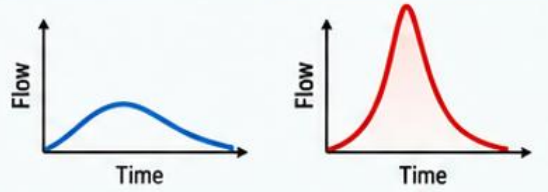
3. Vegetation/Land Cover

Vegetation slows runoff, increases infiltration and storage.



More interception and infiltration
Less runoff

Less interception
More runoff

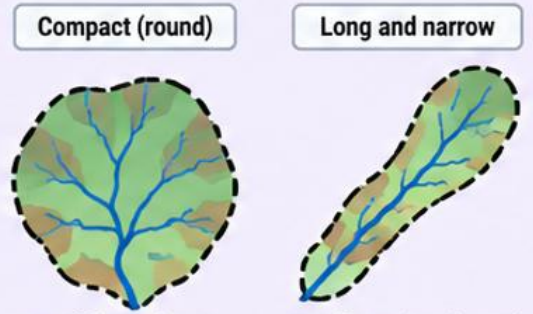


Lower and delayed peak

Higher and earlier peak

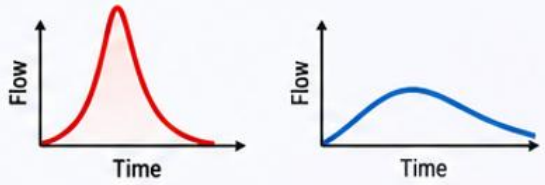
4. Watershed Shape

Watersheds with the same area can produce different hydrographs.



Runoff from all parts reaches the outlet quickly.

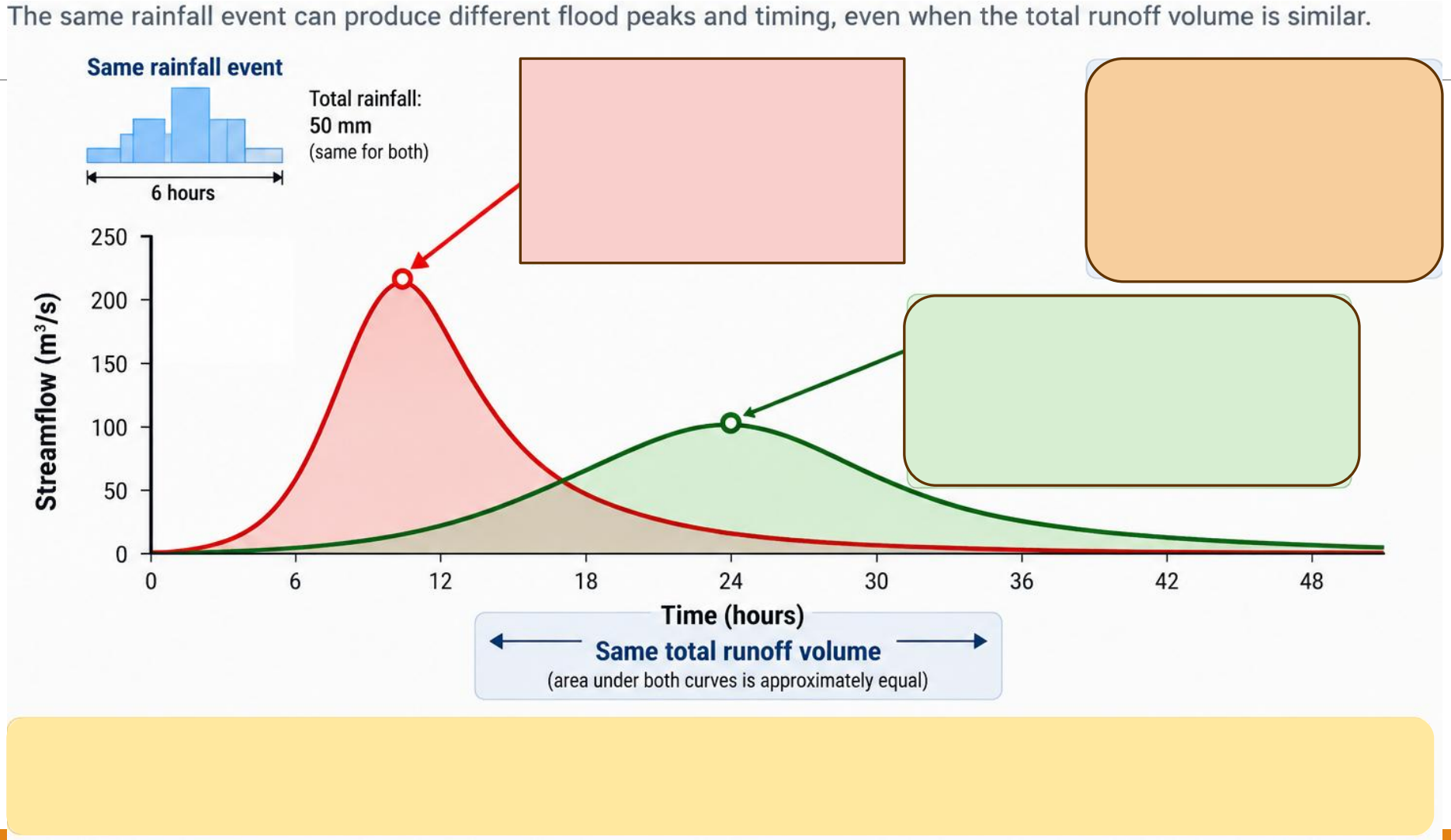
Runoff reaches the outlet at different times.

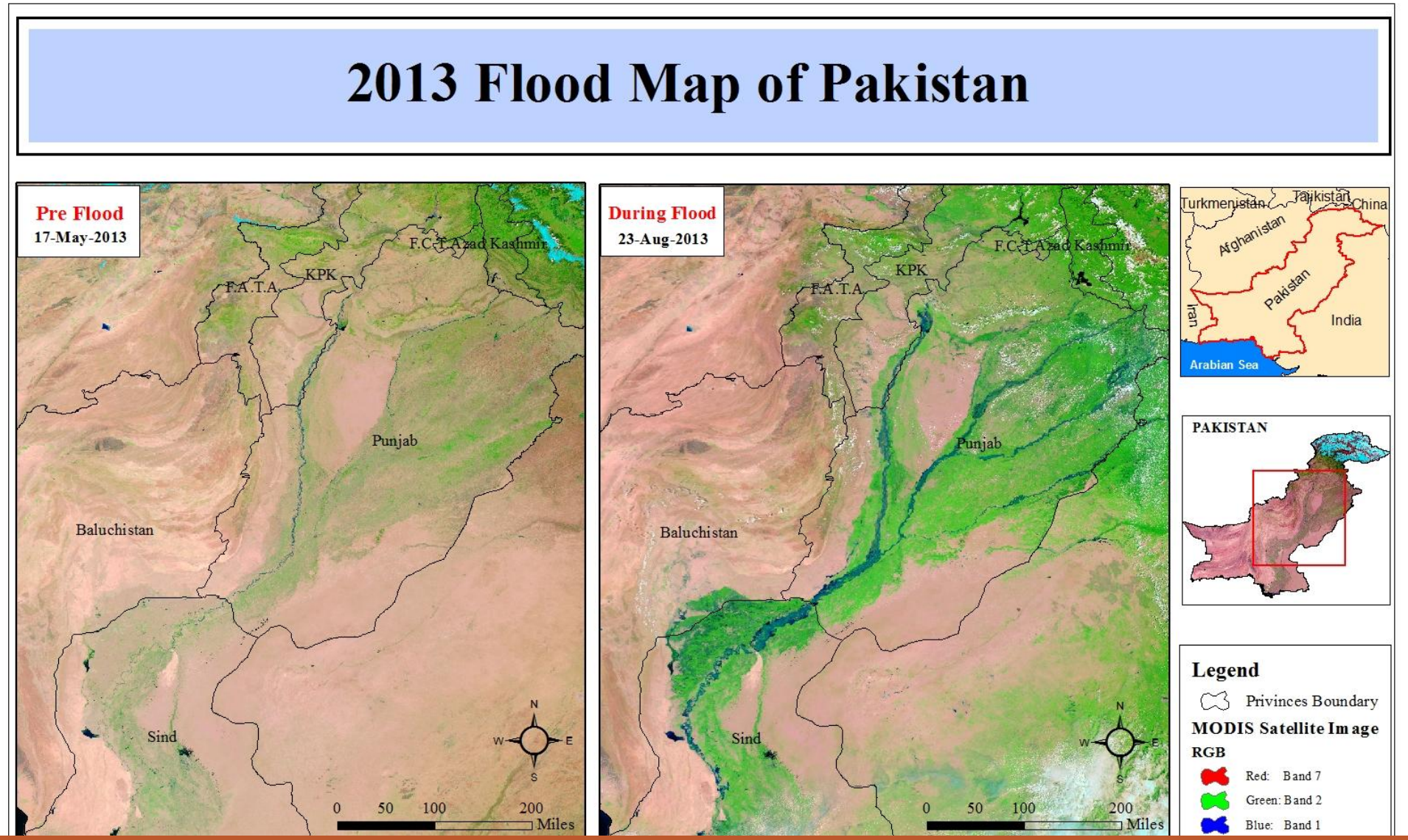


Higher and earlier peak

Lower and later peak

Same monthly rainfall—different runoff risk





HYDROGRAPHS

Floods 2013 in Sindh

FLOOD 2013 SITUATION IN SINDH

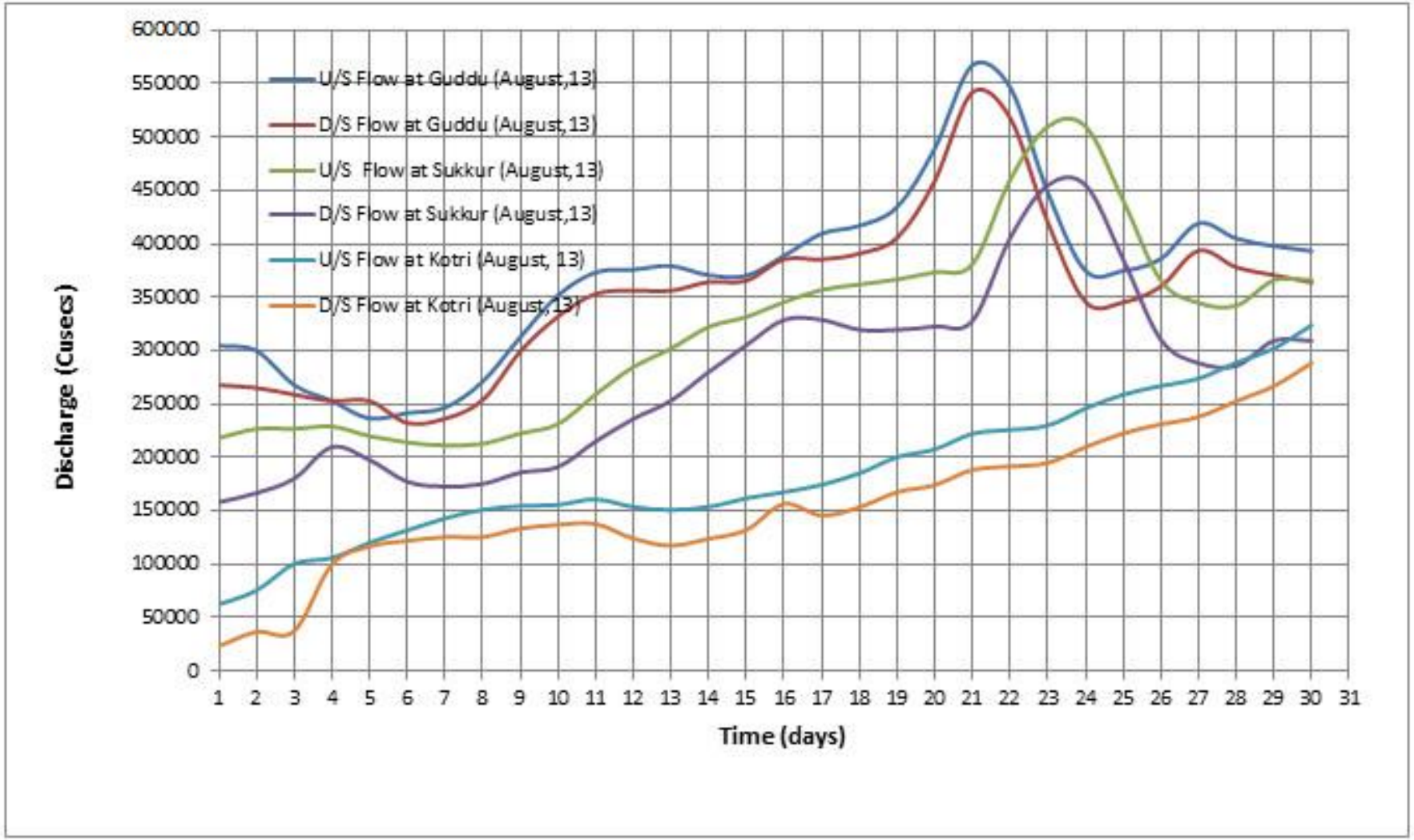


Fig.1: shows Discharge at U/S and D/S of Guddu, Sukkur and Kotri barrages

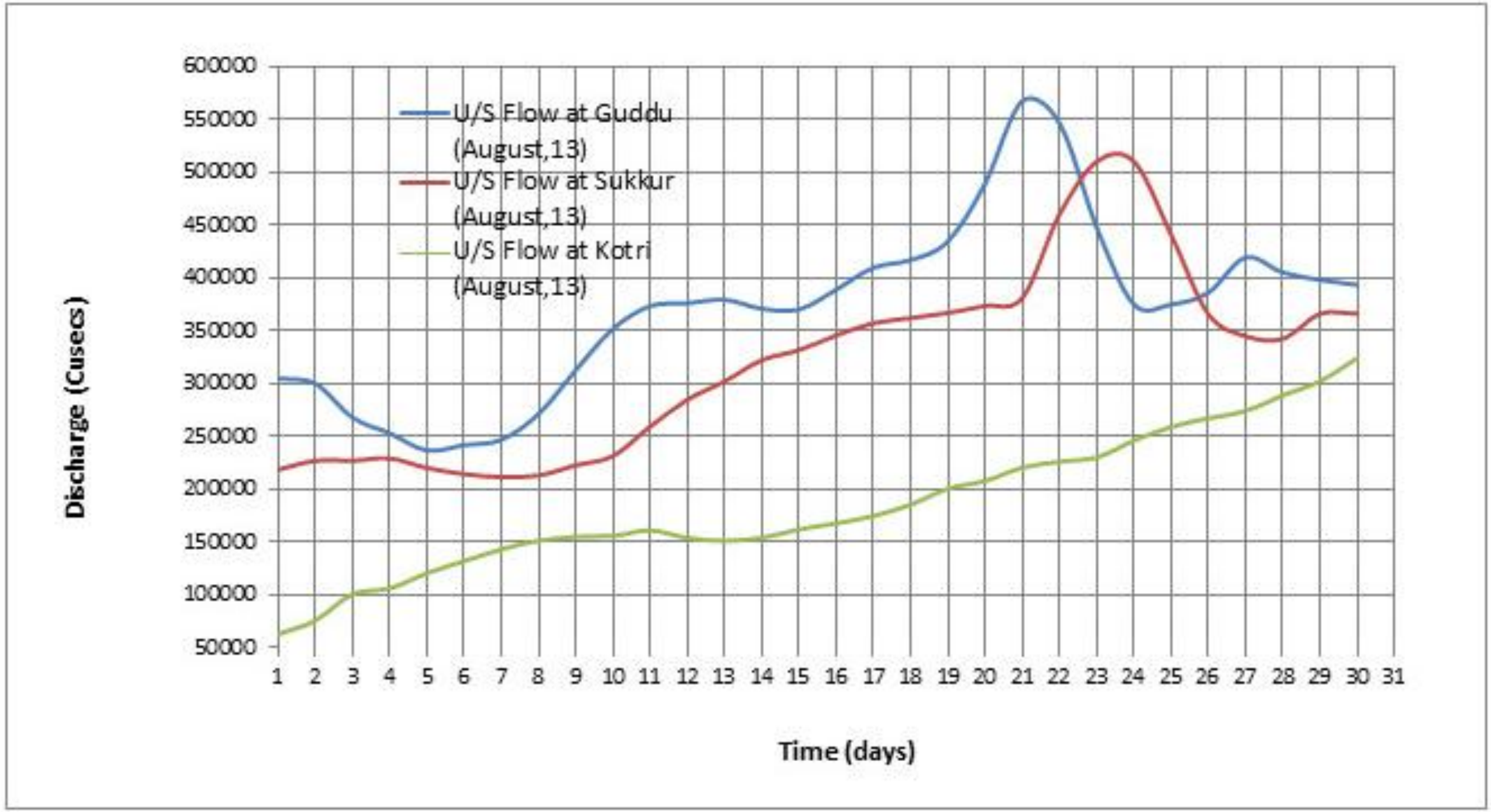


Fig.2: shows Discharge at U/S Guddu , Sukkur and Kotri barrages

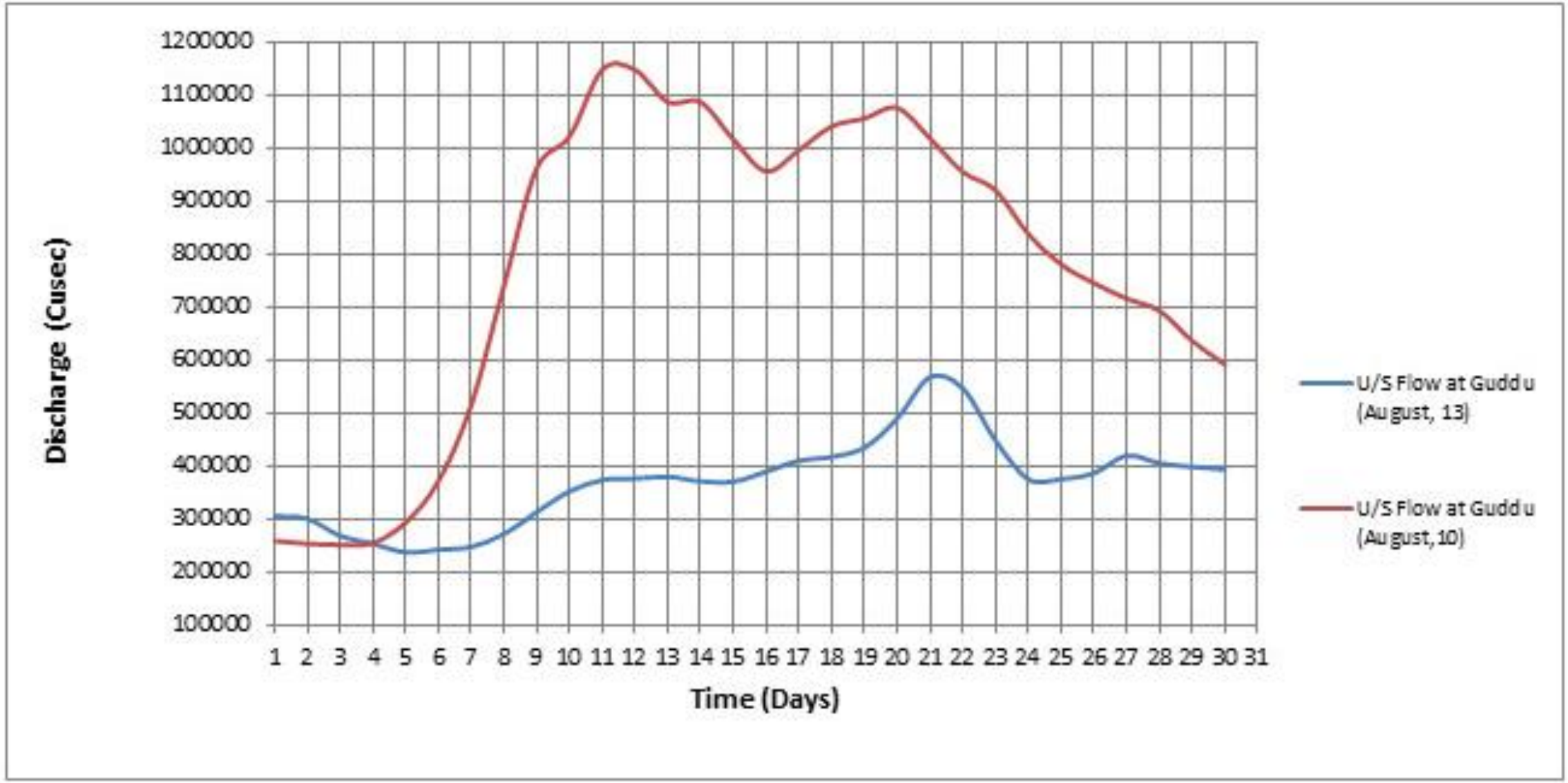


Fig. 4: shows comparison of flood 2010 and 2013 at U/S Guddu

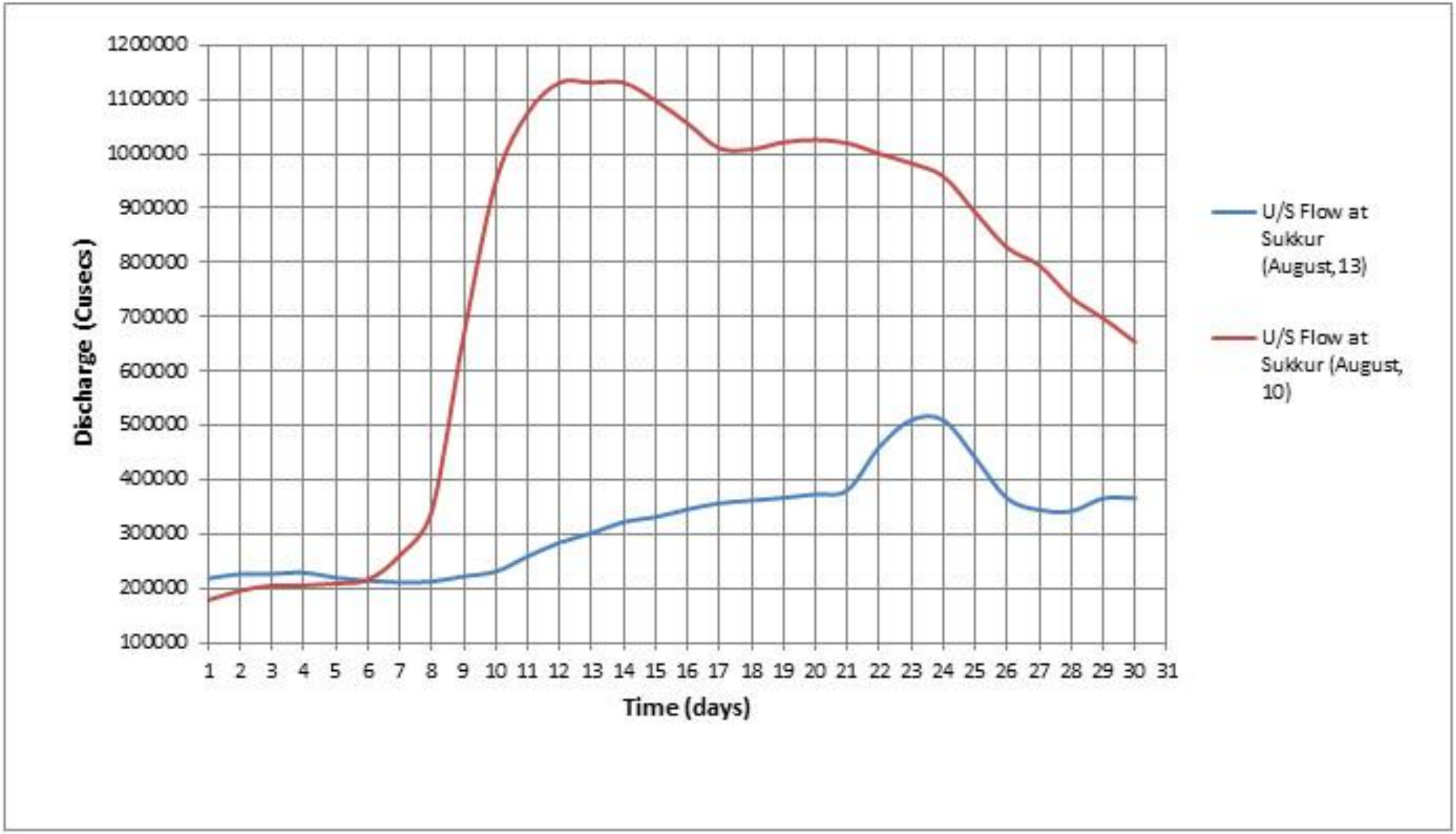
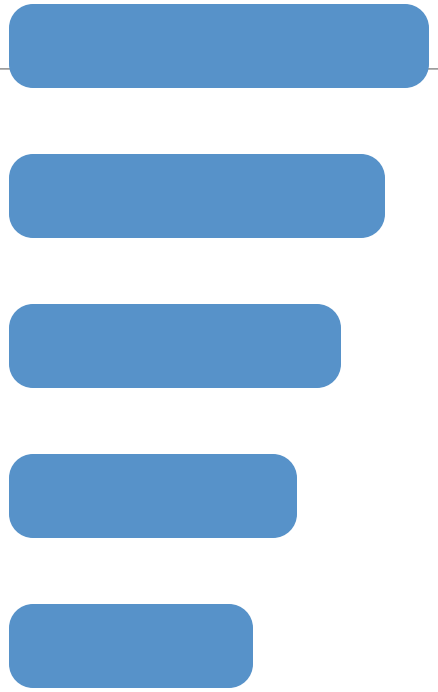


Fig. 6: shows comparison of flood 2010 and 2013 at U/S, Sukkur

From observed rainfall and catchment characteristics to predicted discharge

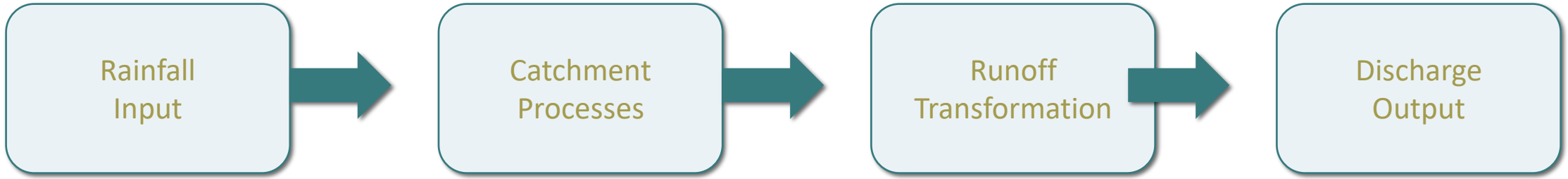


Rainfall–Runoff Modeling Concept

Objective: estimate the runoff or streamflow response produced by a rainfall event or rainfall time series.

- Inputs may include rainfall, soil properties, land cover, topography, and antecedent conditions.
- Observed discharge is used for calibration and evaluation where measurements are available.
- Model choice depends on the purpose, data availability, catchment scale, and required level of detail.

A simple Conceptual Model



Losses and storage control how much rainfall becomes effective rainfall.

Catchment routing controls when that water reaches the outlet.

Gauge and Satellite data are Complements

Gauge strengths

Direct point measurement; long local records where maintained.

Gauge limitations

Sparse coverage; missing values; wind/exposure errors.

Satellite strengths

Spatially complete, near-real-time products.

Satellite limitations

Bias in mountains and intense localized storms; validate locally.

Choosing a rainfall–runoff method

The method should match the decision, watershed scale, and available data.

Rational Method: Estimates peak discharge for suitable small catchments; not event volume.

Curve Number Method: Estimates event runoff depth from rainfall, soil, and land-cover conditions.

Hydrological and Hydraulic (H&H) modeling: Can simulate runoff timing and routing when suitable data are available.

Use local evidence and check assumptions before accepting results.

Regression or ML as a Hydrologic Analysis Tool

- Regression can relate runoff or streamflow to explanatory variables when a suitable data record is available.
- Regression works best for long time periods.
- Possible predictors may include rainfall characteristics, basin descriptors, or other hydrometeorological variables.
- Regression is an empirical relationship; it should be evaluated using appropriate goodness-of-fit and validation measures.

Can we trust the model results?

A good-looking result is not enough without evidence.

Check rainfall depth, timing, and spatial coverage.

Compare simulated and observed flood peaks, timing, and volume.

Test reasonable ranges of uncertain parameters.

Validate with another event or independent field evidence.

Report uncertainty rather than relying on one number.

4. Two families of runoff models

Peak runoff models and continuous runoff models



RUNOFF MODELS

Peak runoff models

- Provide only the estimates of peak discharge from the watershed

Continuous runoff models

- This class of runoff models provides storm hydrographs for a given rainfall hyetograph
- Provide an estimate of runoff vs. time series

5. Peak runoff models — the Rational Method

Assumptions, the peak-discharge formula and the runoff coefficient

PEAK RUNOFF MODELS

Rational Method

NRCS Method

RATIONAL METHOD

To calculate peak runoff from small watersheds

Provides peak runoff rate from a catchment given:

- the runoff coefficient C,
- the time of concentration T_c,
- the area of the catchment, and
- the information to calculate the input or design storm or rainfall event

RATIONAL METHOD: ASSUMPTIONS

- Catchment is small (less than 200 acres)
- Catchment is concentrated
- Rainfall intensity is uniform over the area of study
- The runoff coefficient is catch-all coefficient that incorporates all losses of the catchment

RATIONAL METHOD: FORMULA

$$Qp = CI_iA$$

Qp = Peak discharge (cfs)

C = runoff coefficient (function of soil type, landuse/cover, and watershed slope)

I_i = average rainfall intensity (in/hr) for a storm duration equal to the time of concentration, T_c

A = area (acre)

For runoff coefficient: refer to Bedient Table 6-5, page 381 (more detailed table in McCuen book --will be shared)

TABLE 7–9 Runoff Coefficients for the Rational Formula versus Hydrologic Soil Group (A, B, C, D) and Slope Range

	A			B			C			D		
Land Use	0–2%	2–6%	6%+	0–2%	2–6%	6%+	0–2%	2–6%	6%+	0–2%	2–6%	6%+
Cultivated land	0.08[a]	0.13	0.16	0.11	0.15	0.21	0.14	0.19	0.26	0.18	0.23	0.31
	0.14[b]	0.18	0.22	0.16	0.21	0.28	0.20	0.25	0.34	0.24	0.29	0.41
Pasture	0.12	0.20	0.30	0.18	0.28	0.37	0.24	0.34	0.44	0.30	0.40	0.50
	0.15	0.25	0.37	0.23	0.34	0.45	0.30	0.42	0.52	0.37	0.50	0.62
Meadow	0.10	0.16	0.25	0.14	0.22	0.30	0.20	0.28	0.36	0.24	0.30	0.40
	0.14	0.22	0.30	0.20	0.28	0.37	0.26	0.35	0.44	0.30	0.40	0.50
Forest	0.05	0.08	0.11	0.08	0.11	0.14	0.10	0.13	0.16	0.12	0.16	0.20
	0.08	0.11	0.14	0.10	0.14	0.18	0.12	0.16	0.20	0.15	0.20	0.25
Residential lot	0.25	0.28	0.31	0.27	0.30	0.35	0.30	0.33	0.38	0.33	0.36	0.42
size 1/8 acre	0.33	0.37	0.40	0.35	0.39	0.44	0.38	0.42	0.49	0.41	0.45	0.54
Residential lot	0.22	0.26	0.29	0.24	0.29	0.33	0.27	0.31	0.36	0.30	0.34	0.40
size 1/4 acre	0.30	0.34	0.37	0.33	0.37	0.42	0.36	0.40	0.47	0.38	0.42	0.52
Residential lot	0.19	0.23	0.26	0.22	0.26	0.30	0.25	0.29	0.34	0.28	0.32	0.39
size 1/3 acre	0.28	0.32	0.35	0.30	0.35	0.39	0.33	0.38	0.45	0.36	0.40	0.50
Residential lot	0.16	0.20	0.24	0.19	0.23	0.28	0.22	0.27	0.32	0.26	0.30	0.37
size 1/2 acre	0.25	0.29	0.32	0.28	0.32	0.36	0.31	0.35	0.42	0.34	0.38	0.48
Residential lot	0.14	0.19	0.22	0.17	0.21	0.26	0.20	0.25	0.31	0.24	0.29	0.35
size 1 acre	0.22	0.26	0.29	0.24	0.28	0.34	0.28	0.32	0.40	0.31	0.35	0.46
Industrial	0.67	0.68	0.68	0.68	0.68	0.69	0.68	0.69	0.69	0.69	0.69	0.70
	0.85	0.85	0.86	0.85	0.86	0.86	0.86	0.86	0.87	0.86	0.86	0.88
Commercial	0.71	0.71	0.72	0.71	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72
	0.88	0.88	0.89	0.89	0.89	0.89	0.89	0.89	0.90	0.89	0.89	0.90
Streets	0.70	0.71	0.72	0.71	0.72	0.74	0.72	0.73	0.76	0.73	0.75	0.78
	0.76	0.77	0.79	0.80	0.82	0.84	0.84	0.85	0.89	0.89	0.91	0.95
Open space	0.05	0.10	0.14	0.08	0.13	0.19	0.12	0.17	0.24	0.16	0.21	0.28
	0.11	0.16	0.20	0.14	0.19	0.26	0.18	0.23	0.32	0.22	0.27	0.39
Parking	0.85	0.86	0.87	0.85	0.86	0.87	0.85	0.86	0.87	0.85	0.86	0.87
	0.95	0.96	0.97	0.95	0.96	0.97	0.95	0.96	0.97	0.95	0.96	0.97

[a] Runoff coefficients for storm recurrence intervals less than 25 years.

[b] Runoff coefficients for storm recurrence intervals of 25 years or longer.

Key Details of Table 7-9

Parameters covered:

- **Land Use / Cover Types:** such as agricultural, residential, commercial, industrial, or open/natural land covers like grasslands and forests.
- **Hydrologic Soil Groups (A, B, C, D):** Classifies soils based on infiltration and transmission rates (from high infiltration sand/gravel types in Group A to low infiltration clay types in Group D).
- **Slope Ranges:** Categorized by general gradient classes (e.g., flat, rolling, steep), which affect the velocity and volume of surface runoff.

Application: The table values represent empirical runoff coefficients (C) used in peak discharge equations.

RUNOFF COEFFICIENT FOR NONHOMOGENEOUS AREA

Weighted runoff coefficient based on the area of each land use

$$Cw = \sum_{j=1}^{n} Cj\ Aj / \sum_{j=1}^{n} Aj$$

6. The NRCS Curve Number method

The rainfall–runoff equation, S and initial abstraction

NRCS RUNOFF CURVE NUMBER METHODS

- By the USA Soil Conservation Service (now called the Natural Resources Conservation Service), division of the USDA (US Department of Agriculture)
- To predict peak discharge due to a 24-hr storm event
- Empirically derived relationships that use precipitation, land cover, hydrologic soil group, and watershed characteristics to estimate runoff depth. When combined with a unit hydrograph method, it can also be used to estimate peak discharge and runoff hydrographs.
- More sophisticated approach than the Rational Method

NRCS CURVE NUMBER

Curve number is a coefficient that reduces the total precipitation to runoff potential, after “losses”

- Evaporation
- Absorption
- Transpiration
- Surface Storage

The higher the CN value, the higher the runoff potential will be

It is essential to use the CN value that best mimics the **Ground Cover Type** and **Hydrologic Condition**

NRCS RAINFALL-RUNOFF EQUATION

The following equation presents the relationship between accumulated rainfall and accumulated runoff

$$Q = \frac{(P - I_a)^2}{(P - I_a) + S}$$ Equation 1

Where:

Q = accumulated direct runoff (inches or mm)

P = accumulated rainfall (potential maximum runoff) (inches or mm)

I_a = initial abstraction including surface storage, interception, evaporation, and infiltration prior to any runoff occurring (inches or mm)

S = potential maximum soil moisture retention after runoff begins (inches or mm)

Note: for $P \leq I_a$, $Q = 0$

POTENTIAL MAXIMUM RETENTION (S)

Potential maximum retention (S) can be calculated using Equation 2.

$$S = z\left(\frac{100}{CN} - 1\right)$$

Equation 2

Where:

z=10 for English measurement units, or 254 for metric

CN = Runoff Curve Number (0 - 100)

Generally, I_a may be estimated as $\mathbf{I_a = 0.2\ S}$ Equation 3

Substituting I_a value in Equation 1

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)}$$

Equation 4

NRCS RUNOFF EQUATIONS

$$Q = \frac{(P - I_a)^2}{(P - I_a) + S} \qquad \text{[eq. 2-1]}$$

where

Q = runoff (in)
P = rainfall (in)
S = potential maximum retention after runoff begins (in) and
I_a = initial abstraction (in)

$$I_a = 0.2S \qquad \text{[eq. 2-2]}$$

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)} \qquad \text{[eq. 2-3]}$$

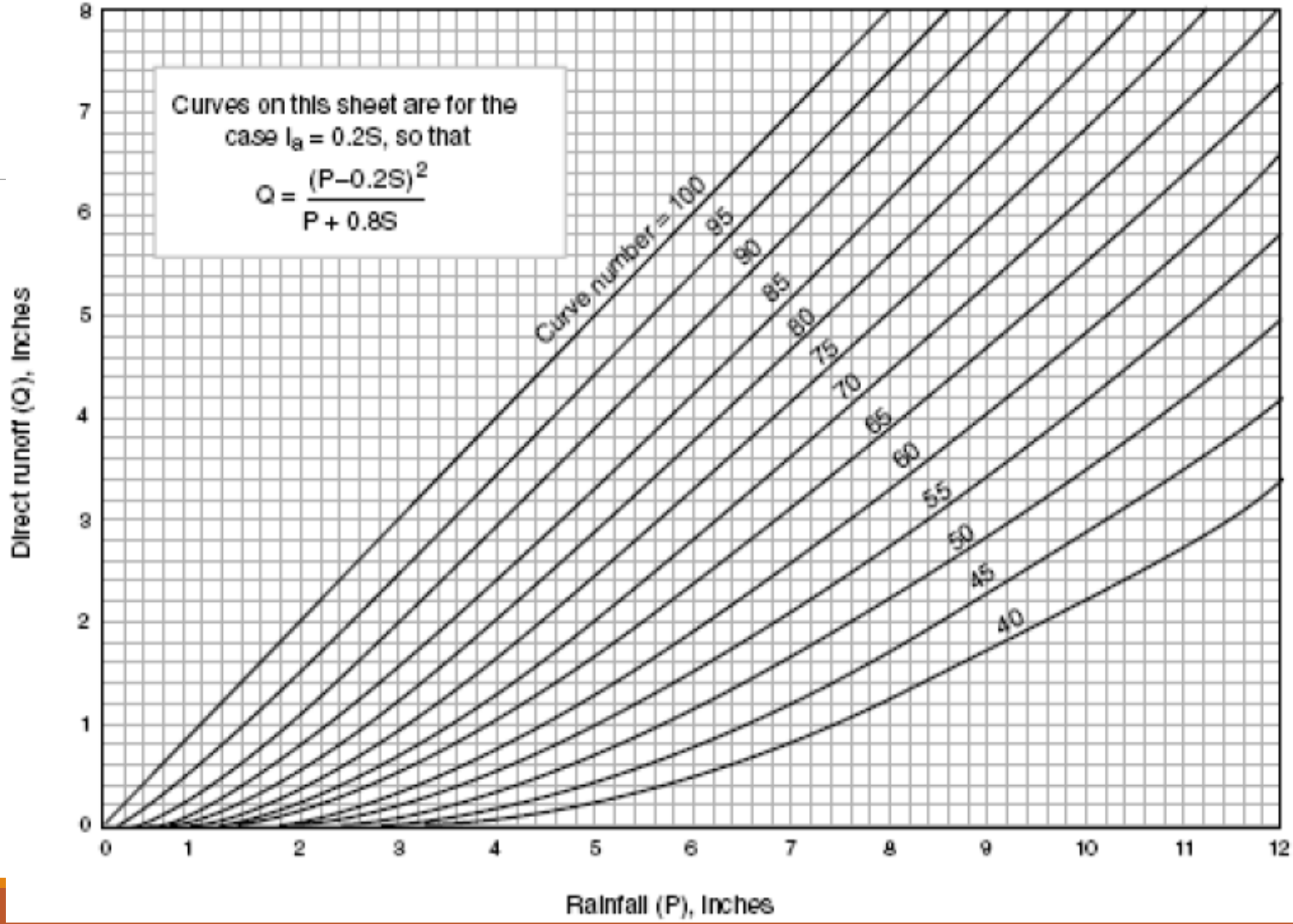
$$S = \frac{1000}{CN} - 10 \qquad \text{[eq. 2-4]}$$

Figure 2-1 (runoff equation):

$$Q = \frac{\left[P - .2\left(\frac{1000}{CN} - 10\right)\right]^2}{P + 0.8\left(\frac{1000}{CN} - 10\right)}$$

where
Q = runoff (in)
P = rainfall (in)
CN = runoff curve number

The solution to this equation results in runoff depth in "watershed" inches.


Curves on this sheet are for the case $I_a = 0.2S$, so that
$Q = \frac{(P-0.2S)^2}{P+0.8S}$
Curve number = 100
95
90
85
80
75
70
65
60
55
50
45
40
Direct runoff (Q), Inches
Rainfall (P), Inches
0
1
2
3
4
5
6
7
8
9
10
11
12

Curve Number CN

The principal physical watershed characteristics affecting the relationship between rainfall and runoff are

- land use, land treatment,
- soil types, and
- land slope.

NRCS method uses a combination of soil conditions and land uses (ground cover) to assign a runoff factor to an area (CN).

CN indicates the runoff potential of an area

- The higher the CN, the higher the runoff potential

Soil properties also influence the relationship between runoff and rainfall since soils have differing rates of infiltration.

Based on infiltration rates, the NRCS has divided soils into four hydrologic soil groups.

HYDROLOGIC SOIL GROUPS

Hydrologic Group is a grouping of soils that have similar runoff potential under similar storm and land cover conditions

Group A Soils: High infiltration (low runoff). Sand, loamy sand, or sandy loam. Infiltration rate > 0.3 inch/hr when wet.

Group B Soils: Moderate infiltration (moderate runoff). Silt loam or loam. Infiltration rate 0.15 to 0.3 inch/hr when wet.

Group C Soils: Low infiltration (moderate to high runoff). Sandy clay loam. Infiltration rate 0.05 to 0.15 inch/hr when wet.

Group D Soils: Very low infiltration (high runoff). Clay loam, silty clay loam, sandy clay, silty clay, or clay. Infiltration rate 0 to 0.05 inch/hr when wet.

Effects of Urbanization: Consider the effects of urbanization on the natural hydrologic soil group. For compacted soil, you should make appropriate changes to the selected soil group.

Antecedent soil moisture conditions: AMC I, II, and III

ANTECEDENT SOIL MOISTURE CONDITIONS-AMC

AMC is the preceding relative moisture of the pervious surfaces prior to the rainfall event

- **Low**: when there has been little preceding rainfall
- **High:** when there has been considerable preceding rainfall prior to the modeled rainfall event

ACM I (dry), ACM II (average), and ACM III (wet)

For modeling purposes, we consider watersheds to be AMC II, which is essentially an average moisture condition

For areas with high water table conditions, AMC III antecedent soil moisture conditions may be used.

CNs

Land Use	Hydrologic Soil Group			
	A	B	C	D
Pre-Developed or Undisturbed Pervious	55	71	80	84
Developed or Disturbed Pervious	71	80	84	88
Impervious	98	98	98	98

A CN of 100 is to be used for permanent water surfaces such as lakes and ponds

Table 2: NRCS runoff curve numbers (CN) for selected urban land use[1]

Cover description		*Curve numbers for hydrologic soil group*			
Cover type and hydrologic condition	**Average impervious area[2]**	**A**	**B**	**C**	**D**
Fully developed urban areas (vegetation established)					
Open space (lawns, parks, golf courses, cemeteries, etc)[3]:					
Poor condition (grass cover <50%)		68	79	86	89
Fair condition (grass cover 50% to 75%)		49	69	79	84
Good condition (grass cover >75%)		39	61	74	80
Impervious areas:					
Paved parking lots, roofs, driveways, etc (excluding ROW)		98	98	98	98
Streets and roads:					
Paved: curbs and storm sewers (excluding ROW)		98	98	98	98
Paved: open ditches (including ROW)		83	89	92	93
Gravel (including ROW)		76	85	89	91
Dirt (including ROW)		72	82	87	89
Western desert urban areas:					
Natural desert landscaping (pervious areas only)[4]		63	77	85	88
Artificial desert landscaping (impervious weed barrier, desert shrub with 1-2 inch sand or gravel mulch and basin borders)		96	96	96	96
Urban districts:					
Commercial and business	85	89	92	94	95
Industrial	72	81	88	91	93
Residential districts by average lot size:					
1/8 acre or less (town houses)	65	77	85	90	92
1/4 acre	38	61	75	83	87
1/3 acre	30	57	72	81	86
1/2 acre	25	54	70	80	85
1 acre	20	51	68	79	84
2 acres	12	46	65	77	82
Developing urban areas					
Newly graded areas (pervious areas only, no vegetation)[5]	77	86	91	94	
Idle lands (CN's are determined using cover types similar to those in Table 3)					

1 Average runoff condition and I_a=0.2S.
2 The average percent impervious area shown was used to develop the composite CN's. Other assumptions are as follows: impervious areas are directly connected to the drainage system, impervious areas have a CN of 98, and pervious areas are considered equivalent to open space in good hydrologic condition. CN's for other combinations of conditions pay be computed using Figures 3 or 4.
3 CN's shown are equivalent to those of pasture. Composite CN's may be computed for other combinations of open space cover type.
4 Composite CN's for natural desert landscaping should be computed using Figures 3 or 4, based on the impervious area percentage (CN=98) and the pervious area CN. The pervious area CN's are assumed equivalent to desert shrub in poor hydrologic condition.
5 Composite CN's to use for the design of temporary measures during grading and construction should be computed using Figures 3 or 4 based on the degree of development (impervious area percentage) and the CN's for the newly graded pervious areas.

COMPOSITE CURVE NUMBER

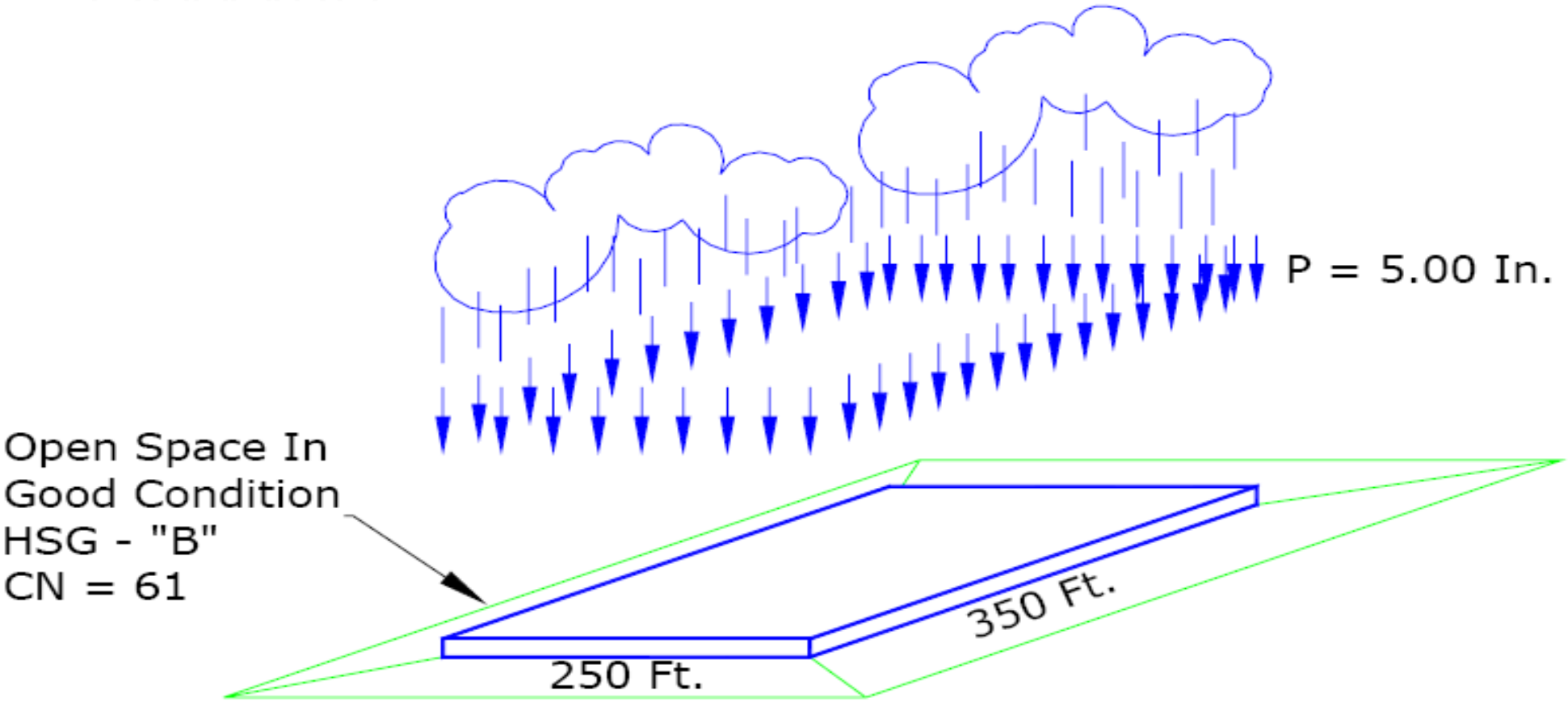
When a drainage area has more than one land use

When Composite CN is used

- The drainage area is considered a uniform land use represented by the composite curve number
- can be calculated by using the weighted method

Composite Curve Number Calculation Example			
Land use	% of total land area	Curve number	Weighted curve number (% area x CN)
Residential 1/8-acre soil group B	0.80	85	68
Meadow good condition soil group C	0.20	71	14
Total Weighted CN = 68 + 14 = 82			

EXAMPLE



How much runoff volume would be expected from this circumstance ?

Runoff Volume example (Continued):

1. Compute the Surface Storage:
   $S = (1000 / CN) - 10$
   $S = (1000 / 61) - 10 = 6.393$ Inches
2. Compute the Initial Abstraction:
   $I_a = 0.2 \times S$
   $I_a = 0.2 \times 6.393 = 1.279$ Inches
3. Compute the runoff in Watershed Inches:
   $Q = (P - I_a)^2 / (P - Ia + S)$
   $Q = (5.00 - 1.279)^2 / (5.00 - 1.279 + 6.393)$
   Q = **1.369 Inches** ***(Remember the original P=5.00 Inches)***
4. Compute the Runoff Volume:
   V = [1.369 In / (12 In / Ft)] x 250 Ft x 350 Ft =
   V = **9983 CF** or 9983 CF / (43560 SF / Ac) = **0.2293 Ac-Ft**

V_{PRECIP} = [5.000 In / (12 In / Ft)] x 250 Ft x 350 Ft = 36,458 CF
Therefore the CN reduced the Precipitation Volume by 75%!

7. Continuous runoff models

The time-area method and unit hydrograph techniques

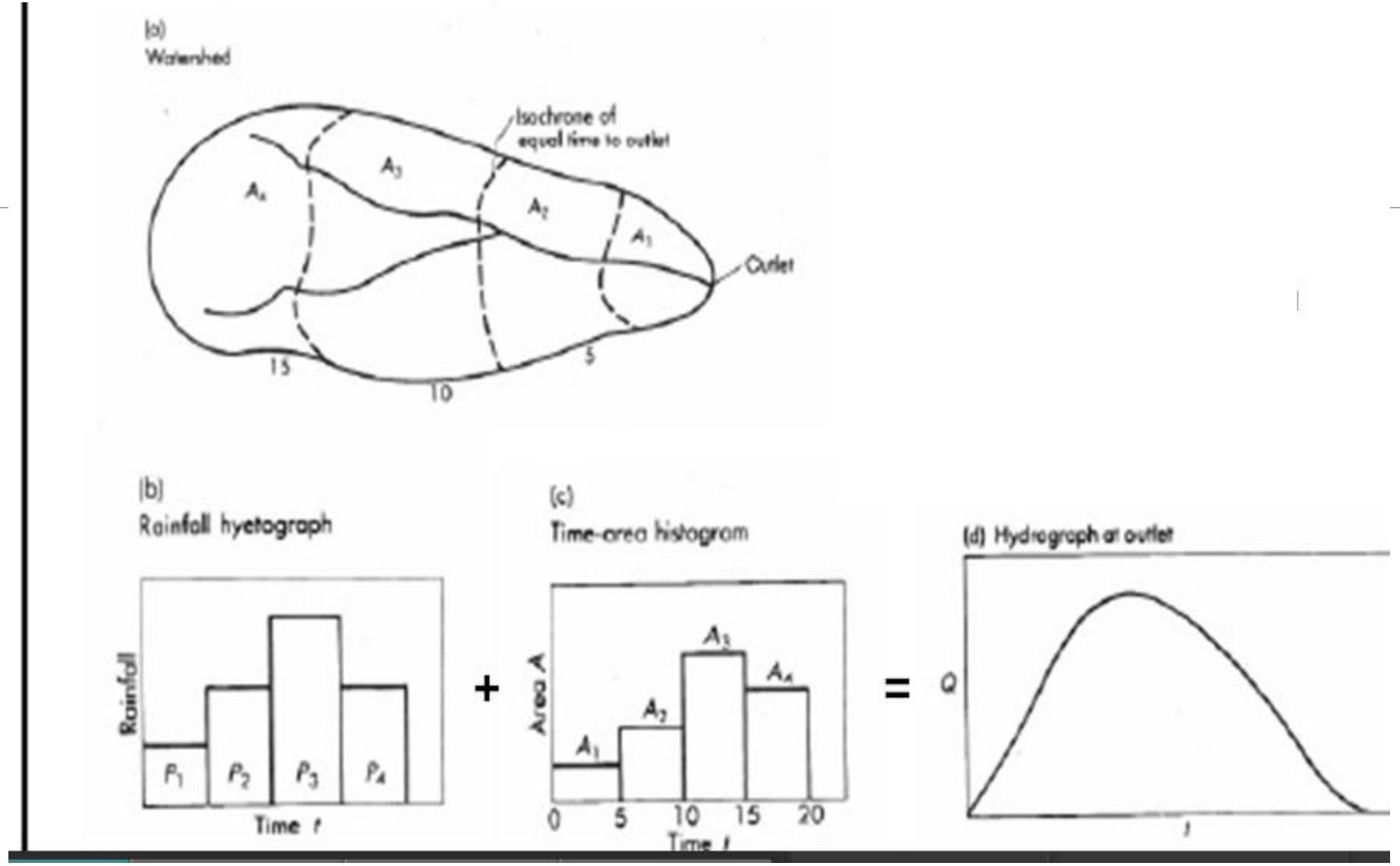
CONTINUOUS RUNOFF MODELS

Time area method

Unit Hydrograph Techniques

TIME AREA METHOD

- Develop to address non-uniform rainfall in large areas
- Convert rainfall excess into a hydrograph
- Concept of time-area histogram is used
- This method assumes that outflow hydrograph results from pure translation of direct runoff to the outlet at uniform velocity, ignoring any storage effects in the watershed
- Watershed is divided into subareas with distinct runoff translation times to the outlet
- Subareas are delineated with **isochrones of equal translation time** (numbered upstream from the outlet)


(a)
Watershed
Isochrone of
equal time to outlet
A_4
A_3
A_2
A_1
Outlet
15
10
5
(b)
Rainfall hyetograph
Rainfall
P_1
P_2
P_3
P_4
Time t
(c)
Time-area histogram
Area A
A_1
A_2
A_3
A_4
0
5
10
15
20
Time t
(d) Hydrograph at outlet
Q
t

KEY TAKEAWAYS

- Only part of the rainfall becomes runoff: effective rainfall = gross rainfall − losses, and runoff is generated when rainfall exceeds the soil's infiltration capacity.
- A catchment transforms rainfall input into streamflow output — losses and storage control how much becomes runoff, and routing controls when it arrives.
- The hydrograph (rising limb, peak, recession) records the catchment response; the area under the curve is the runoff volume, and its shape reflects both storm and catchment characteristics.

KEY TAKEAWAYS (CONTINUED)

- The runoff coefficient C captures soil, land use/cover and slope; a non-homogeneous area needs a weighted C by area.
- The NRCS Curve Number method estimates event runoff depth from $Q = (P - 0.2S)^2 / (P + 0.8S)$, with $S = (z / CN) - z$.
- The higher the CN, the higher the runoff potential; CN depends on land use, soil (hydrologic soil group A–D), slope and antecedent moisture (AMC I/II/III).
- Multiple land uses are combined into a composite CN;

IN-CLASS ACTIVITY

- A catchment has CN = 80 and receives P = 5.0 inches under average moisture conditions (AMC II).
- Compute the potential maximum retention: $S = (1000 / CN) - 10$.
- Compute the initial abstraction: $Ia = 0.2\ S$.
- Compute the direct runoff: $Q = (P - 0.2S)^2 / (P + 0.8S)$.
- Discuss how Q would change for a higher CN (say 90) and for a wetter antecedent condition (AMC III).

QUIZ: RUNOFF MODELS

- 1. Effective rainfall is best described as:
- 2. When does excess water begin to contribute to runoff?
- 3. On a hydrograph, streamflow is made up of:
- 4. On a hydrograph, the area under the curve represents:
- 5. Compared with a gently sloping catchment, a steep catchment generally produces:
- 6. Two watersheds receive the same rainfall but

QUIZ: RUNOFF MODELS (CONTINUED)

- 9. The Rational Method formula is:
- 10. The Rational Method assumes the catchment is:
- 11. For a non-homogeneous catchment, the weighted runoff coefficient is computed by:
- 12. A higher curve number (CN) means:
- 13. For CN = 75 (English units), the potential maximum retention S is:
- 14. Which hydrologic soil group has the highest infiltration rate?

Thank You